

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-5

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

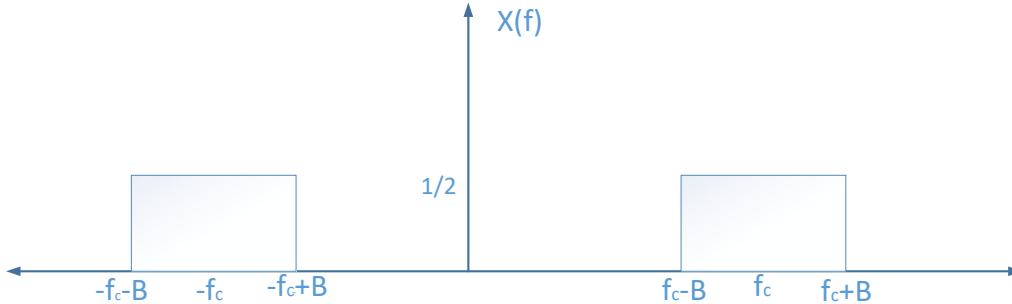
Date: 10-09-2025

Objectives

- SSB Modulation, AM Generation Circuits.

Tutorial Problems

1. A DSB-SC modulated signal at the carrier frequency f_c in the frequency domain is shown in figure below. Find the USSB in the time domain.



2. A message signal $m_1(t)$ of bandwidth 10 Hz amplitude modulates a carrier signal using the balanced ring modulator, as shown in the figure. Find the signal at the output of the bandpass filter.

